

Softpro School of AI — 45-Day AI Development Summer Training

Course code: SSAI-101 **Author:** Arjit Verma **Duration:** 45 days (~9 weeks, Mon–Sat) **Contact hours:** ~200 (5 hrs/day × 40 working days) **Target price:** ₹15,000/seat **Cohort size:** 50/batch (2 parallel batches planned)

1. Who This Is For

- Pre-final / final year **B.Tech (CS/IT/ECE)**
- **BCA, MCA, M.Sc (CS/IT)** students
- **Early-career professionals** (0–2 years experience) looking to pivot into AI
- All years of engineering welcome (with pre-work completion)

Prerequisite: Python basics. Checked and polished in the free pre-work week.

2. Outcomes (ranked)

1. **A job or internship offer** — curriculum designed backward from hiring manager requirements
2. **A strong portfolio** — 3 deployed projects + 1 capstone, all public on GitHub with demo videos
3. **Certificate + AI skills on resume**
4. **Skills to build your own AI product**

3. Design Principles

- **Everything ships.** Every project is deployed to a public URL. No localhost-only code.
 - **Free-tier first.** Entire course can be completed without spending a single rupee on APIs or infra. Students build real things on free cloud tiers and local models.
 - **Framework-agnostic.** Students see LangChain, LangGraph, CrewAI, AutoGen, OpenClaw, and Llama Index — and learn when to pick which.
 - **Multi-provider LLMs.** Gemini, Groq (Llama/Mixtral), Ollama (local), Hugging Face. Students don't get locked into one vendor's ecosystem.
 - **Placement-aligned.** Final week is resume + LinkedIn + GitHub + mock interviews + portfolio launch.
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4. The Free-Tier Stack

Every tool chosen so students can complete the course and keep building after, at ₹0/month.

Layer	Tool	Free-tier limit
Primary LLM	Google Gemini API (Flash 2.0)	1,500 requests/day, very generous
Fast inference	Groq API (Llama 3, Mixtral)	Free tier, blazing fast
Local LLM	Ollama (Llama 3.1, Phi-3, Qwen)	Free forever, runs on laptop
Model hub	Hugging Face Inference API	Free tier for hosted models
Embeddings	sentence-transformers (local) + Gemini embeddings (free)	₹0
Vector DB	Chroma (local) + pgvector via Supabase (free tier)	500MB Supabase free
Database	Supabase PostgreSQL	500MB free
UI	Streamlit + Gradio	Free, Python-native
Deployment	Hugging Face Spaces, Streamlit Community Cloud, Render free tier	₹0
Observability	Langfuse (open-source, self-host or free cloud)	Free
Code host	GitHub	Free
Learning LMS	Softpro Handshake (later: Softpro School of AI LMS)	Included

Students spend ₹0 on infra during the course. This is a major selling point — "learn AI development without a credit card."

5. Pre-Work Week (free, self-paced, before Day 1)

Sent 7 days before batch starts. ~15-20 hours total.

- **Python polish:** functions, OOP, decorators, async basics, type hints, Pydantic
- **Tooling setup:** VS Code + extensions, terminal comfort, Git + GitHub, Python venv / uv
- **Accounts to create** (all free): Gemini API key, Groq API key, Hugging Face, Supabase, Render, GitHub, Streamlit Community Cloud
- **Ollama installation** on student's laptop + first local model (Phi-3 or Llama 3.1 8B)

- **Pre-assessment quiz** — 30 questions, auto-graded with feedback
- **Welcome call** — 1-hour Zoom with trainer on Day 0

Students who skip pre-work are flagged on Day 1 and given remedial support in Week 1.

6. Curriculum — Day by Day

Phase 1 — Foundations (Days 1-10)

Goal: Everyone at baseline. First working LLM app by Day 2.

Day	Topic
1	Python power-up for AI: requests, async, dotenv, Pydantic. First Gemini API call.
2	LLM fundamentals: tokens, context windows, sampling (temperature, top-p), model families
3	Prompt engineering I: system prompts, zero-shot, few-shot, chain of thought
4	Prompt engineering II: structured outputs with Pydantic, JSON mode
5	Multi-provider patterns: switching between Gemini, Groq, Ollama with a simple abstraction
6	Working with data: JSON, CSV, PDF (PyPDF2/pdfplumber), lightweight scraping (BeautifulSoup)
7	Streamlit crash course — building LLM UIs in one afternoon
8	Error handling, retries, rate limits, cost awareness
9	Mini-project 1 build day
10	Deploy (HF Spaces / Streamlit Cloud) + demo day

Mini-project 1: AI-powered content summarizer Input: PDF, article URL, or YouTube transcript. Output: structured summary (key points, TL;DR, action items). Stack: Streamlit + Gemini + Python. Deployed free on Hugging Face Spaces or Streamlit Community Cloud.

Phase 2 — RAG & Memory (Days 11-20)

Goal: Build apps that use external knowledge — the single most employable AI skill right now.

Day	Topic
11	Embeddings fundamentals: what they are, free options (sentence-transformers, Gemini embeddings)
12	Semantic search from scratch (numpy + cosine similarity, no framework)
13	Chroma (local, free) — first vector DB, persistence, collections
14	pgvector via Supabase — moving from local Chroma to cloud-hosted free Postgres
15	Chunking strategies: fixed, recursive, semantic, parent-document
16	Retrieval: k-NN, hybrid search (BM25 + vector)
17	Re-ranking (free options: sentence-transformers cross-encoders)
18	Llama Index introduction — when it beats raw code
19	Mini-project 2 build day
20	Deploy + demo

Mini-project 2: Document Q&A System Input: upload college notes, textbooks, or research papers. Output: ask natural-language questions, get cited answers. Stack: Streamlit + Chroma or pgvector + Gemini + Llama Index. Deployed free on HF Spaces.

Phase 3 — Agents & Tools (Days 21-32)

Goal: Go from Q&A bots to agents that take actions. This is the differentiator.

Day	Topic
21	Tool use / function calling — first principles, no framework
22	LangChain basics — chains, memory, the 20% you actually need
23	LangGraph — state machines for AI, when it makes sense
24	Building a ReAct agent with LangChain (research + summarize tool)
25	CrewAI — multi-agent made simple, role-based agents
26	AutoGen (Microsoft) — conversation-based multi-agent pattern
27	OpenClaw — lightweight agentic framework
28	Framework comparison workshop — build the same agent in 3 frameworks, pick which fits
29	Observability: Langfuse self-hosted, tracing an agent, debugging failures
30	Guardrails & safety: prompt injection defense, output validation
31	Mini-project 3 build day
32	Deploy + demo

Mini-project 3: Personal research agent Goal: an agent that researches a topic across the web, takes notes, and produces a briefing. Stack: student picks their framework (LangGraph / CrewAI / AutoGen). Must use 3+ tools. Traced in Langfuse. Deployed free.

Phase 4 — Capstone + Placement (Days 33-45)

Goal: Ship one portfolio-grade project + get placement-ready.

Day	Activity
33	Capstone ideation workshop + scoping
34	1:1 architecture review with trainer
35-41	Build sprint (7 days) — daily standups, trainer check-ins
42	Deployment day — CI/CD basics, env vars, monitoring
43	Demo day + peer code review
44	Resume workshop + LinkedIn + GitHub polish + mock interviews
45	Portfolio site launch + certificate ceremony

Capstone Options

Student picks one. Must be deployed and demonstrable.

- 1. **Multi-agent research assistant** — topic in, full briefing out, with citations
- 2. **WhatsApp AI business assistant** — Twilio (free trial) + Gemini + Supabase
- 3. **AI customer support for a local business** — RAG on their FAQ/catalog
- 4. **Code review agent** — runs on a GitHub repo, posts PR comments
- 5. **Educational tutor for one subject** — adaptive, personalized study buddy
- 6. **AI finance tracker** — ingests statements, natural-language insights
- 7. **AI recipe generator** — dietary-aware, grocery-list output
- 8. **AI news briefing agent** — daily personalized digest, scheduled

7. Assessments & Certificate

Component	Weight	Notes
Attendance	85%+ required	Gating criteria
Weekly quizzes (6)	20%	Must pass each
Mini-projects (3)	30%	Rubric-graded: code, deployment, docs, demo
Capstone	40%	Rubric + demo day presentation
Peer review participation	10%	Code review sessions

Certificate criteria

- ≥85% attendance
 - All 3 mini-projects submitted and deployed
 - Capstone deployed + documented + demo video
 - Passed final assessment
 - Portfolio site live
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8. Deliverables — What Every Student Walks Out With

- **3 deployed projects** (content summarizer, document Q&A, research agent)
 - **1 capstone project** (deployed, documented, demoed)
 - **GitHub profile** with all 4 repos, polished READMEs, pinned
 - **Portfolio site** on Vercel (free), all projects linked
 - **4 demo videos** (one per project, 2 min each — Loom or YouTube)
 - **LinkedIn** optimized + 4 project-announcement posts
 - **Certificate** from Softpro School of AI
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9. Class Format (daily)

Block	Duration	Activity
Concept + live coding	2 hrs	Trainer-led, questions welcome
Guided build	1 hr	Students code along, TA floats
Independent work	1.5 hrs	Mini-project work + doubt solving
Daily standup	30 min	Progress + blockers + next-day prep

10. Trainer & TA Plan

- **Lead trainer:** Arjit (Year 1, both batches)
- **Teaching assistant:** 1 hire by Month 2 (for parallel batches) — code review, grading, doubt solving

- **Guest lectures:** industry practitioners from Arjit's network (EY, Capgemini, TVS Next alumni) for weekly inspiration sessions
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11. Gear & Infra Requirements

Teaching hall (Softpro-provided):

- High-lumen projector or 65"+ screen
- Stable 100+ Mbps WiFi (non-negotiable — live API demos)
- Air conditioning (Lucknow summer heat)
- Whiteboard + markers
- Wireless presenter + clip-on mic for recording

Per-student (they bring):

- Laptop with 8GB+ RAM (16GB ideal for Ollama)
- Internet at home for homework

Per-student cost to Softpro School of AI:

- ~₹0 in API credits (we're using free tiers)
 - Printing materials, certificate, snacks: ~₹500/student
 - LMS access, cloud backup: ~₹300/student
 - **Total ~₹800/student** — very clean margins
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12. Open Questions for Launch

1. **Batch dates** — first batch starts when? Summer 2026: May/June window?
2. **AKTU credit transfer** — can Softpro's MoU with AKTU make this count as academic credit? Huge enrollment lever.
3. **Payment model** — upfront? 2-installment? EMI via EduFin / GrayQuest?
4. **Refund policy** — 3- or 7-day money-back window recommended for trust
5. **Marketing channels** — @codewitharjit (183K+), Softpro's AKTU channel, college visits, Instagram Ads, partner colleges
6. **Hybrid delivery** — record and stream to students from MoU colleges outside Lucknow? Multiplies revenue at zero marginal cost
7. **Corporate add-on** — can this curriculum also be sold as 45-day corporate reskilling at ₹3L/batch for partner companies?

v1 — iterate after first pilot batch.